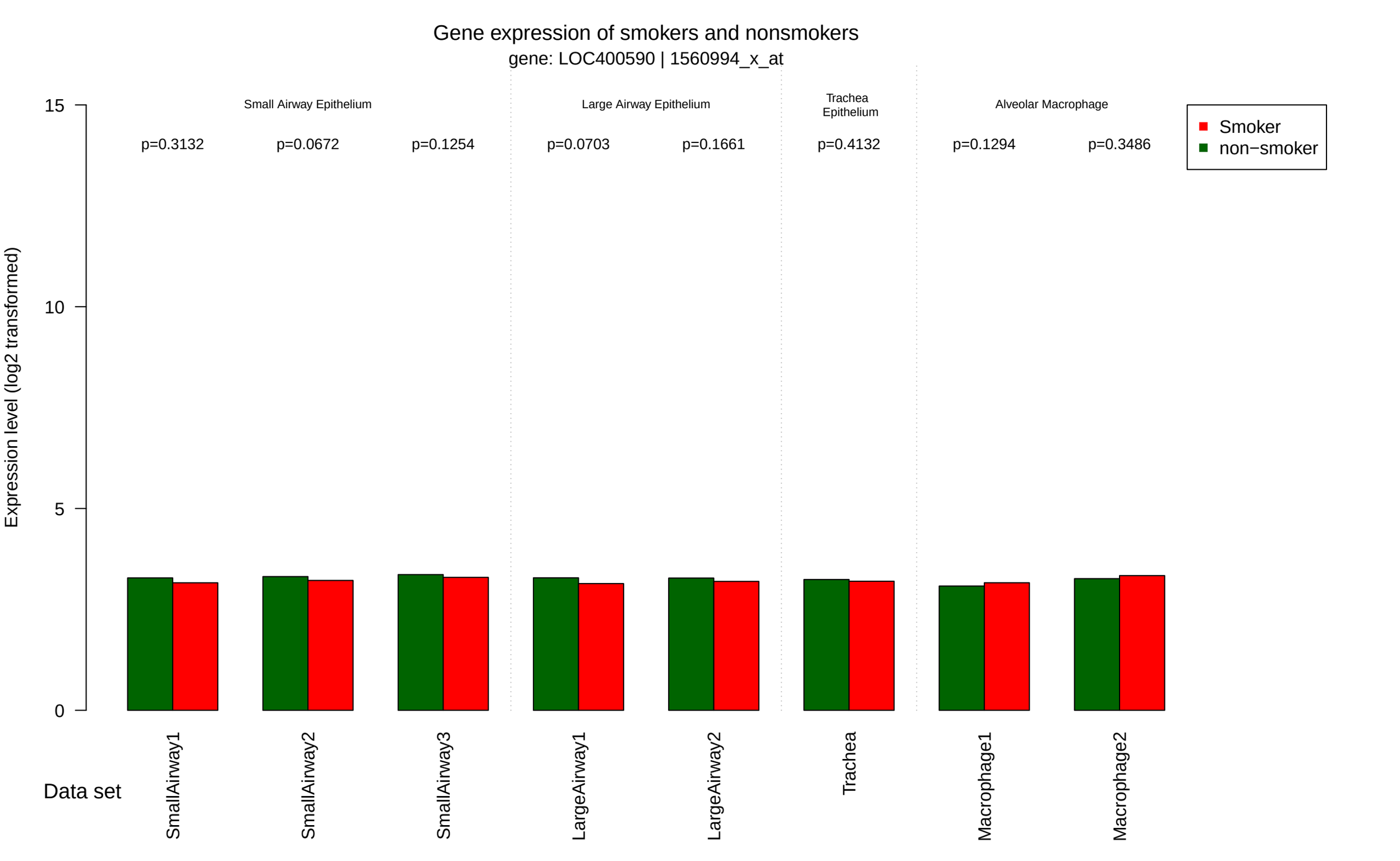
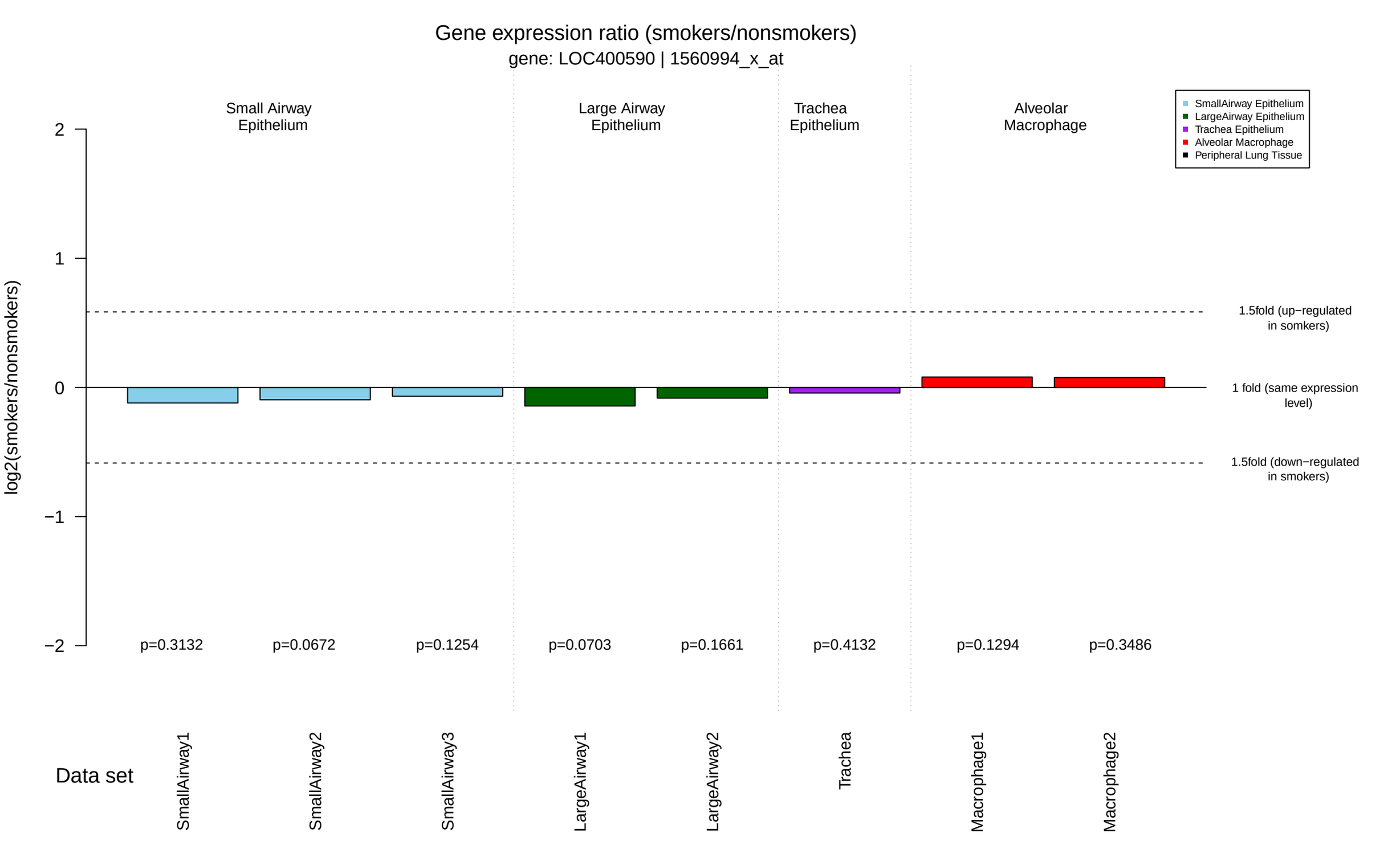
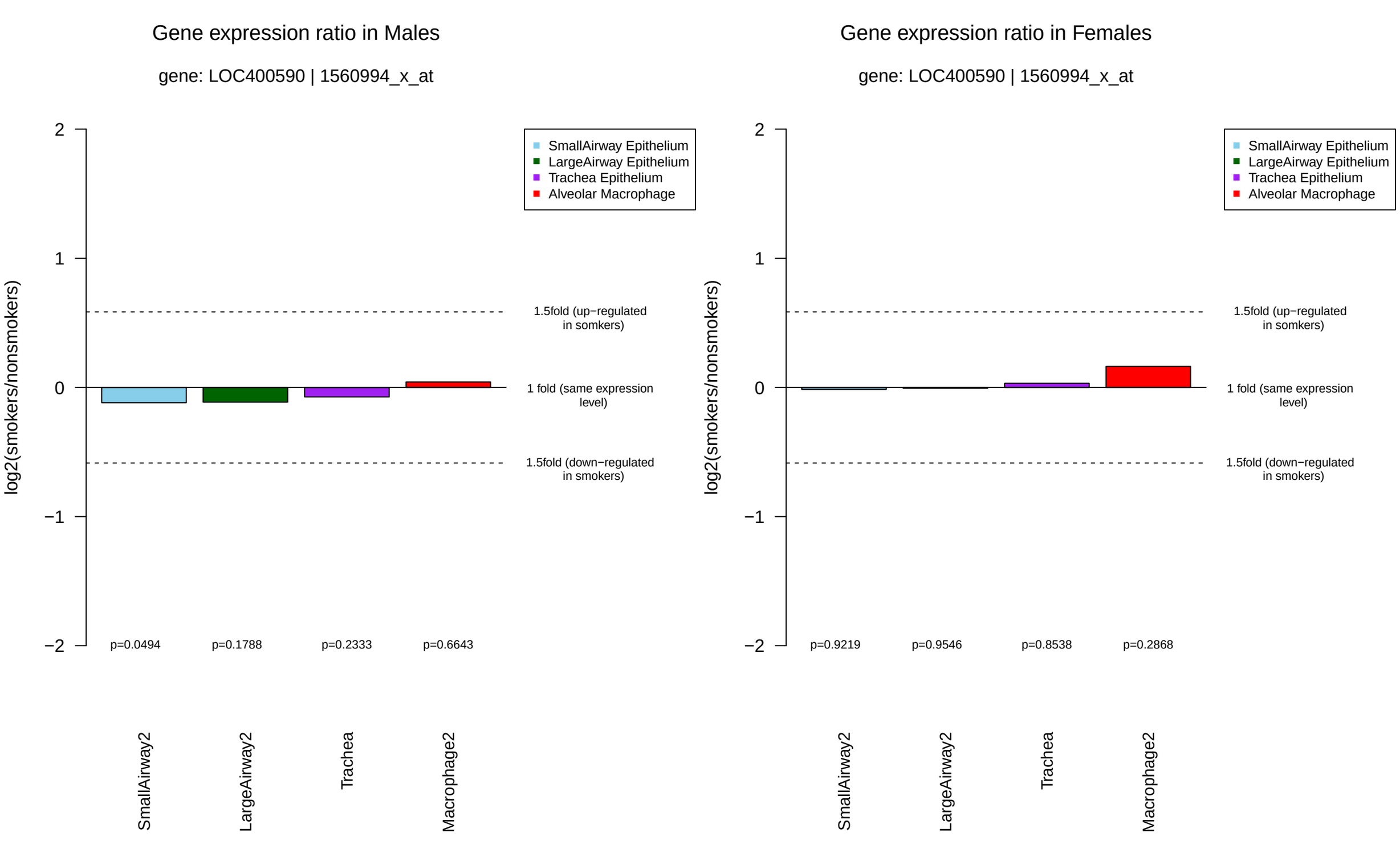
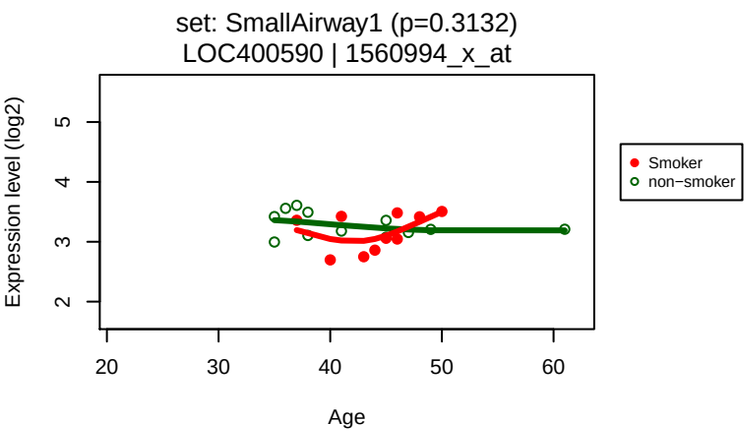
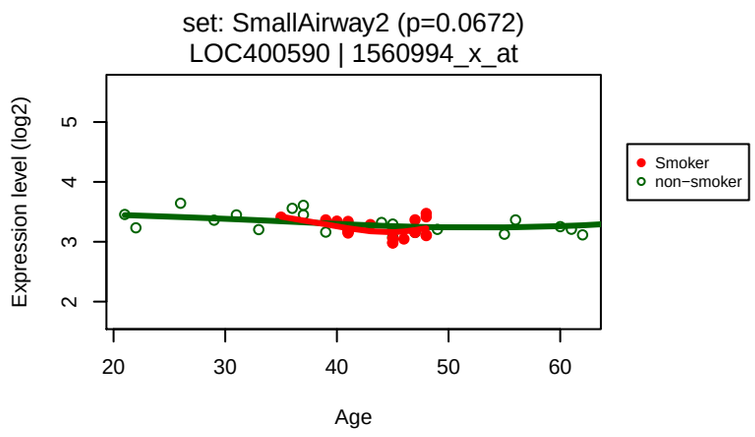
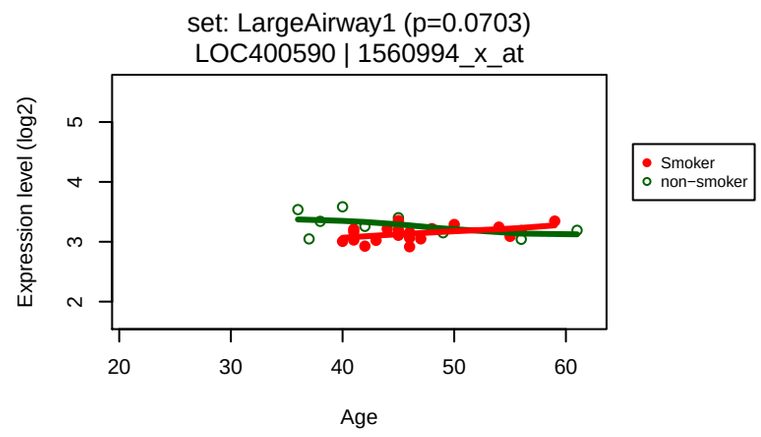

Gene expression of smokers and nonsmokers
gene: LOC400590 | 1560994_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3132
p=0.0672
p=0.1254
p=0.0703
p=0.1661
p=0.4132
p=0.1294
p=0.3486
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

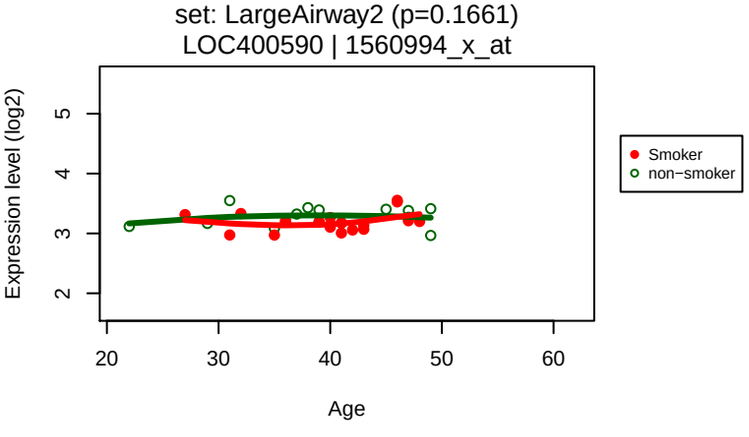

Gene expression ratio (smokers/nonsmokers)
gene: LOC400590 | 1560994_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3132
p=0.0672
p=0.1254
p=0.0703
p=0.1661
p=0.4132
p=0.1294
p=0.3486
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

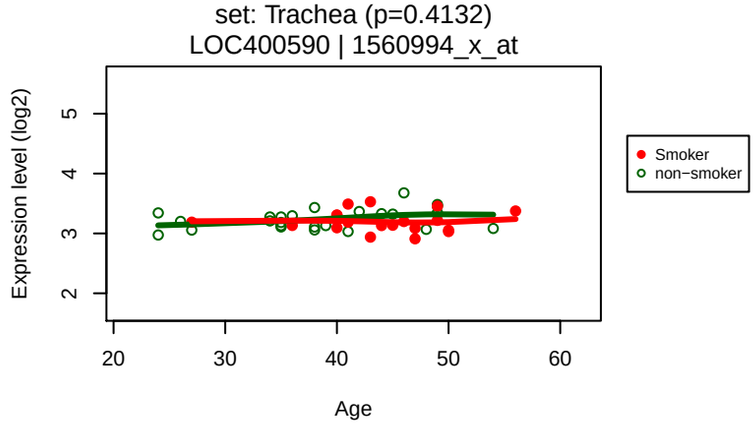

Gene expression ratio in Males
gene: LOC400590 | 1560994_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0494
p=0.1788
p=0.2333
p=0.6643
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LOC400590 | 1560994_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9219
p=0.9546
p=0.8538
p=0.2868
SmallAirway2
LargeAirway2
Trachea
Macrophage2

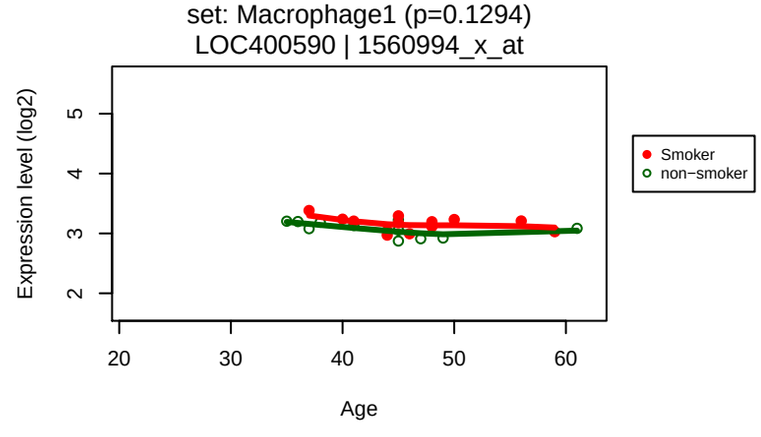

set: SmallAirway1 (p=0.3132)
LOC400590 | 1560994_x_at
Expression level (log2)
Age
Smoker
non−smoker

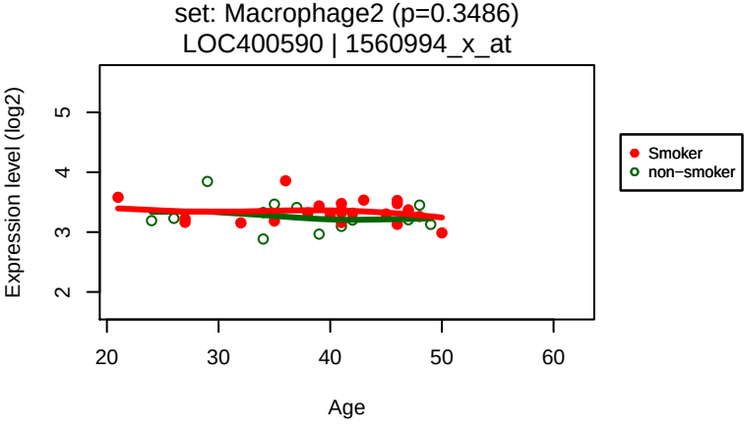

set: SmallAirway2 (p=0.0672)
LOC400590 | 1560994_x_at
Expression level (log2)
Age
Smoker
non−smoker

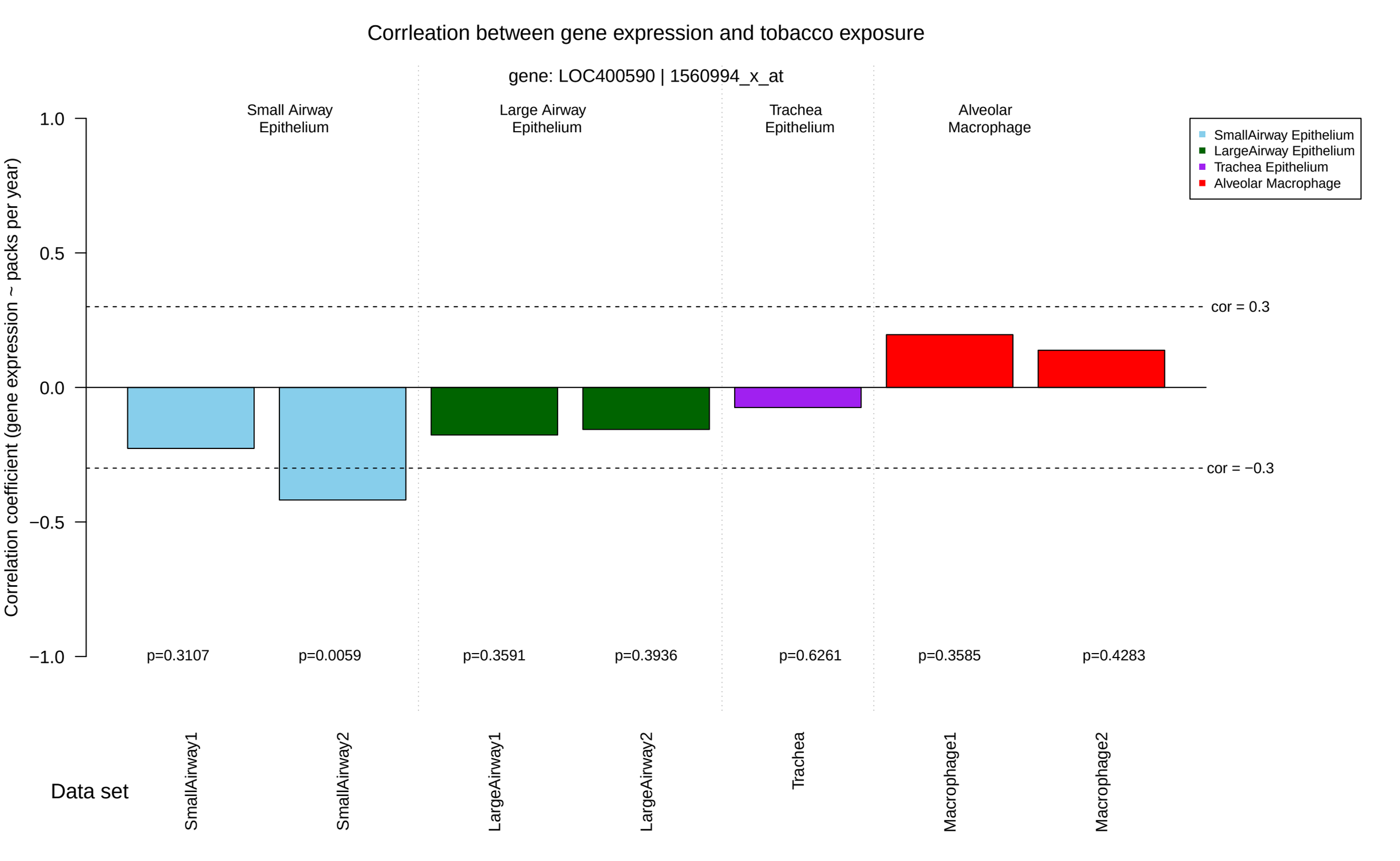

set: LargeAirway1 (p=0.0703)
LOC400590 | 1560994_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1661)
LOC400590 | 1560994_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4132)
LOC400590 | 1560994_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1294)
LOC400590 | 1560994_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3486)
LOC400590 | 1560994_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LOC400590 | 1560994_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3107
p=0.0059
p=0.3591
p=0.3936
p=0.6261
p=0.3585
p=0.4283
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2